**Reflection Journal**

Understanding the function behind the code steps of the machine learning workflow helps demystify what were once stressful, overwhelming commands. Each part of the workflow loading data, fitting the model, testing, and others has a role to play to keep the data clean. Loading and holding back data from the test set for the first time is equivalent to the real world case. The goal here is to determine whether the model is generalizing and not memorizing.

In supervised learning, a student learns using an answer key. The inputs (features matrix (X)) and the target (vector (y)) are given to the model during the training process to determine the rules and the mapping function. In the testing phase, the targets are removed to evaluate the model using only the learned mapping function. Unsupervised learning models learn the structure of the data and cluster them, since there is no target provided.

Decision trees have a lower variance when using ensembles, or the Random Forest approach. The downside of individual decision trees is the variance and their tendency to fit and overfit noise. With Random Forests, many different trees are grown on many different, smaller data sets and feature sets, resulting in trees with variance that cancels each other out, making for a much more stable and accurate model.

In addition to all of the previous tasks, interpreting ensemble model results helps reveal new domain-specific insights. For the Iris classification task, it was found that nearly all predictions were made using petal measurements and only a very small number of decisions were based on

the sepal measurements. It was shown that statistical findings correspond to the world of botany in the fact that petal sizes tend to vary a lot between different species, showing that machine learning is less about creating syntax for code and more about finding the trends and patterns.